C programing

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DEP:ECE-F

WEEK 13

```c
/*
 * Complete the 'balancedSum' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts INTEGER_ARRAY arr as parameter.
 */

int balancedSum(int arr_count, int* arr)
{
    int totalsum = 0;
    for (int i =0;i<arr_count;i++){
        totalsum += arr[i];
    }
    int leftsum =0;
    for(int i =0;i<arr_count;i++){
        int rightsum = totalsum - leftsum -arr[i];
        if(leftsum==rightsum){
            return i;
        }
        leftsum +=arr[i];
    }
    return 1;

}

```

	Test	Expected	Got	
✓	`int arr[] =  {1,2,3,3};` `printf("%d", balancedSum(4, arr))`	2	2	✓

Passed all tests! ✓

```c
/*
 * Complete the 'arraySum' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts INTEGER_ARRAY numbers as parameter.
 */

int arraySum(int numbers_count, int *numbers)
{
    int sum =0;
    for (int i =0;i<numbers_count;i++){
        sum = sum+numbers[i];
    }
    return sum;
}
```

	Test	Expected	Got	
✓	`int arr[] = {1,2,3,4,5};` `printf("%d", arraySum(5, arr))`	15	15	✓

Passed all tests! ✓

```c
/*
 * Complete the 'minDiff' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts INTEGER_ARRAY arr as parameter.
 */
#include <stdlib.h>
int compare(const void *a, const void *b){
    return (*(int*)a - *(int*)b);
}
int minDiff(int arr_count, int* arr)
{
    qsort(arr, arr_count,sizeof(int), compare);
    int totaldiff=0;
    for(int i =1;i<arr_count;i++){
        totaldiff += abs(arr[i]-arr[i-1]);
    }
    return totaldiff;
}

```

	Test	Expected	Got	
✓	int arr[] = {5, 1, 3, 7, 3}; printf("%d", minDiff(5, arr))	6	6	✓

Passed all tests! ✓